

# Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 3: Seed Dormancy: Why Do Some Healthy-Looking Seeds Refuse to Germinate?

Teacher reference only. Do not project or hand out.

## Quick marking grid

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 |
|----|----|----|----|----|----|----|
| C  | B  | D  | C  | A  | A  | D  |

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share, introduction of the word 'dormancy'

**Q1. In Trans Nzoia, a farmer plants maize and beans on the same day after rain. Some healthy-looking seeds never germinate. Which statement best describes seed dormancy?**

- A. A state in which the seed has died and cannot grow again
- B. A state in which the seed has no embryo inside it
- C. A resting state in which a living seed does not germinate even when conditions look right**
- D. A state in which every seed germinates immediately when watered

**Correct answer: C.** Dormancy is a resting state of a living seed that does not germinate even when conditions seem suitable. The seed is not dead.

USE AFTER: OBSERVE PHASE — Part A: examining seed specimens and comparing dry and soaked seeds

**Q2. Learners soaked hard-coated Senna seeds in water for several hours. The seeds did not swell. Which is the most likely reason?**

- A. The embryo was too large to take up water
- B. The hard seed coat did not let water enter the seed**
- C. The seeds had absorbed too much water already
- D. Soaking always kills seeds that have hard coats

**Correct answer: B.** If a seed does not swell after soaking, water is not entering. A hard, impermeable seed coat blocks water uptake and so keeps the seed dormant.

USE AFTER: OBSERVE PHASE — Part B: setting up four labelled containers, including the control

**Q3. In a dormancy-breaking investigation, one container holds untreated seeds kept moist in the same place as the treated seeds. Why is this container needed?**

- A. To give the treated seeds extra water
- B. To make sure all the seeds germinate
- C. To show the seeds in the other containers are not dormant
- D. To compare with the treated seeds, so we can tell whether the treatment made a difference**

**Correct answer: D.** The control is untreated but kept under the same conditions. Any difference between it and the treated seeds can then be linked to the treatment.

USE AFTER: OBSERVE PHASE — Part B: scarification container (seed coat rubbed on sandpaper or rough stone)

**Q4. A group wants to break dormancy in Acacia seeds by scarification. Which action is scarification?**

- A. Washing the seeds in several changes of water
- B. Leaving the seeds in dry soil for a month
- C. Rubbing the seed coat lightly on sandpaper or a rough stone**
- D. Keeping the seeds in a cool place for some days

**Correct answer: C.** Scarification means scratching, nicking or rubbing the seed coat so water and gases can enter. Washing is leaching and cool storage is chilling.

USE AFTER: EXPLAIN PHASE — Claim-Evidence-Reasoning: causes of dormancy and ways of breaking it

**Q5. A farmer finds that some healthy seeds contain chemical inhibitors that stop germination. Which treatment would best help remove the inhibitors?**

- A. Washing the seeds in several changes of water (leaching)**
- B. Drying the seeds in the sun for a week
- C. Putting the seeds in boiling water
- D. Rubbing the seed coat on sandpaper only

**Correct answer: A.** Leaching washes water-soluble inhibitors out of the seed. Sandpaper only opens the coat, and boiling water can kill the embryo.

USE AFTER: EXPLAIN PHASE — Claim-Evidence-Reasoning: reasoning linking water uptake to the embryo

**Q6. After the seed coat of a dormant bean seed is rubbed and the seed is placed in wet cotton wool, the seed swells. What is this uptake of water called, and why is it important for germination?**

- A. Imbibition; it activates enzymes in the embryo**
- B. Scarification; it makes the seed coat thicker
- C. Leaching; it washes inhibitors out of the embryo
- D. Stratification; it keeps the embryo cold and inactive

**Correct answer: A.** Water uptake by a seed is called imbibition. It activates enzymes in the embryo so germination can begin.

USE AFTER: END OF LESSON — Explain: significance of dormancy for plants in Kenyan ecosystems (Acacia in Tsavo)

**Q7. Acacia seeds in the Tsavo drylands can stay dormant for a long time. How does dormancy help these plants survive?**

- A. It makes the seeds germinate faster than other plants' seeds
- B. It removes the need for water when the seeds finally germinate
- C. It makes every seed in the soil germinate on the first rain
- D. It stops seeds from germinating in a short dry spell, and it spreads germination over time**

**Correct answer: D.** Dormancy lets seeds wait for adequate rain and spreads germination over time. Not all seedlings are then lost to one dry spell.